AI in AR/VR: Enhancing Perception, Personalization, and Generative Experiences

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***Abstract*— Artificial Intelligence (AI) plays an essential role in creating immersive experiences for Augmented Reality (AR) and Virtual Reality (VR) applications. The study examines the role of AI in improving perception and personalization and generating content through real-world AR/VR applications including Snapchat facial filters and IKEA Place interior planner and OpenAI's SORA video generator. The study evaluates technical trade-offs between AI architectures (e.g., YOLOv5 vs. ViT), identifies system-level challenges like latency and bias, and reviews privacy preserving methods aligned with GDPR. The paper establishes a framework to evaluate AI-XR intersection through scholarly research combined with industry applications that emphasizes ethical approaches for building reproducible and performance-aware immersive systems.**

I. Introduction

Artificial Intelligence (AI), Augmented Reality (AR), and Virtual Reality (VR) are converging to revolutionize how users interact with digital content.*AI refers to the ability of machines to replicate cognitive functions such as perception, decision-making, and learning* [1]. *AR overlays digital information onto real-world views, while VR immerses users entirely within a simulated environment*. What was futuristic a decade ago is now propelling mainstream uses that combine perception, interactivity, and personalization.

Modern platforms demonstrate this shift clearly. Snapchat employs real-time facial recognition and object tracking to create dynamic filters and effects. IKEA Place uses AR combined with AI-based spatial detection to recommend and preview furniture layouts in real-time. Meanwhile, OpenAI's SORA introduces generative video capabilities that push the boundaries of content creation. These systems not only showcase technological advancements but also signal a deeper trend: AI is enabling smarter, more human-centered immersive experiences.

This paper investigates how AI facilitates innovation in AR/VR through real-world applications and scholarly views. The study discusses the role of AI in three aspects that counts perception enhancement, personalization, and
generative content and identifies open issues including real-time performance, privacy, and ethics. Understanding these dynamics is crucial as Extended Reality (XR) (*an umbrella term encompassing AR, VR, and Mixed Reality*) systems become increasingly intelligent and integral to everyday digital life.

AI, AR, and VR are blending to create more natural and interactive digital experiences. AI helps systems understand their surroundings, learn from users, and even generate content. While AR adds layers to the real world, VR places users in entirely digital spaces. From Snapchat's dynamic filters to IKEA Place's smart furniture placement, and SORA's text to video creation, AI is powering the next wave of immersive tech. This paper reviews how AI supports perception, personalization, and content creation, while also addressing the challenges and ethical considerations that come with it.

TABLE I: Feature Comparison of Immersive Technologies

Aspect	AR	VR	MR	AI
Reality	Mixed	Virtual	Hybrid	Digital
Device	Phone	Headset	HoloLens	Any
Immersion	Low	High	Medium	N/A
Control	Gesture	Controller	Voice/Gesture	Logic/Data
Usage	Navigation	Gaming	Training	Decision

II. Related Work

The surge in AI applications within AR/VR spans academia and industry alike. Wang et al. [37] dissected AI's role in real-time object tracking, Freitas et al. [38] highlighted AI powered VR education, and Khan et al. [39] tackled digital identity protection in immersive realms. Complementing these are insights from IEEE VR [40] and other top conferences emphasizing human-system interactions.

Wang et al. [2] provided a systematic review of how AI supports immersive analytics, highlighting how computer vision and deep learning models enable spatial scene understanding and object recognition in AR/VR interfaces. Similarly, Liu and Deng [4] examined how AI algorithms optimize interior design through AR, mirroring commercial efforts like IKEA Place.

In the context of content generation, Freitas et al. [13] discussed the role of generative models in educational VR applications, revealing opportunities and limitations in real-time synthesis.The growing trend that SORA by OpenAI aimed to follow - which, yet, due to its presence outside the purview of formal IEEE literature, remains largely unexploited in practice-is for intelligent media creation.

Digital identity and personalization have also seen AI-driven advancements. Khan et al. [35] introduced a blockchain-based framework for identity management in the Metaverse, underscoring the growing intersection between AI, AR/VR, and decentralized systems. Gesture recognition and emotion-driven avatar adaptation are additional areas of focus, as explored by Gupta et al. [36] in a study on cartoonification via AI-based gesture tracking.

At the heart of these works lies an affirmation that AI not only enhances the realism and responsiveness of AR/VR but also brings with it issues of latency, bias, and ethics—questions that are still the subject of academic inquiry and applied research.

AI in AR/VR has gained major attention in both academia and industry. Wang et al. [37] explored how AI models support real-time analysis and object tracking. Freitas et al. [38] discussed the role of AI in virtual education, and Khan et al. [39] examined identity protection in immersive environments. To reduce dependence on a single author group, we also reference work from IEEE VR [40], CHI, and ISMAR conferences, which focus on human interaction and system performance in mixed reality settings.

III. AI Framework in AR/VR

A. Perception Enhancement

Snapchat's filters owe their magic to convolutional neural networks (CNNs) that detect facial landmarks faster than you can say "selfie." These AI wizards track smiles, blinks, and cheeky winks in real-time, painting virtual masks and effects that stick like glue. Similarly, IKEA Place uses AI to "see" your room's layout, lighting, and floor planes, letting you virtually drop in that perfect couch without lifting a finger.

For instance, Snapchat filters rely on *convolutional neural networks (CNNs)* and facial keypoint detection to track expressions, orientation, and depth with high accuracy. These filters are dynamically applied based on real-time visual input, made possible by AI models trained on vast facial datasets. Gupta et al. [36] explored a similar architecture for AI-driven gesture recognition and visual transformations.

IKEA Place also uses ARCore and AI algorithms to detect floor planes, lighting, and room geometry, respectively. According to Liu and Deng [4], deep learning models can detect interior layouts and recommend object placements by analyzing spatial constraints. These techniques mirror the perception pipeline used in consumer-grade AR applications.

Now, applying perception contributed by AI, AR/VR systems contextualize information, supplementing the digital overlays into naturalness with the physical environment. This function is essential with respect to realism, user comfort, and accuracy when it comes to immersive interactions.

B. Personalization and Adaptation

The term "personalization" describes how AI customizes immersive experiences for each user. This covers identity management, environment setup, and adaptive content in AR/VR.

For instance, IKEA Place makes recommendations for furniture based on factors like room size, layout, and even user preferences over time. These recommendation systems often use *Reinforcement Learning (RL)* and user-behavior modeling, as reflected in Wang et al. [2], who highlight how AI adapts interfaces in real-time to user interaction patterns.

In the Metaverse, personalization becomes identity-centered. Khan et al. [35] proposed a decentralized identity system using *Non-Fungible Tokens (NFTs)* and blockchain to let users manage their virtual presence securely. This research represents ongoing efforts to provide users with AI-assisted control over avatars, privacy settings, and interactions in immersive environments, even though these elements are not yet common in commercial platforms.

AI acts like a savvy companion, learning your preferences, adjusting the environment, and serving up experiences tailored just for you. Whether it's IKEA Place suggesting a chair that fits your quirky nook or Metaverse avatars reflecting your moods and moves, personalization breathes life into static digital worlds.

C. Content Generation and Synthesis

Generative content is one of the most revolutionary uses of AI in AR/VR. The ability of large-scale, multimodal AI to produce dynamic, immersive video environments from basic prompts is demonstrated by models such as OpenAI's SORA.

Academic parallels exist in Freitas et al. [13], who explored AI-generated environments for educational VR, enabling simulations to evolve in real-time based on learner behavior. Similarly, Wang et al. [2] showed how text-to-scene generation and multimodal synthesis could support immersive analytics in enterprise AR systems.

Generative AI breaks down creative barriers, no more wrestling with 3D models or lines of code. Whether you're an artist or a casual user, tools like OpenAI's SORA let you conjure immersive scenes from simple prompts, putting powerful content creation in everyone's hands.

TABLE II: Comparison of AI Use in AR/VR Platforms

Platform	AI Use Case	Impact
Snapchat	Face/Gesture Tracking	Social interaction
IKEA Place	Interior Planning	Personalized shopping
SORA	Text-to-Video Gen.	Generative media
Metaverse Avatars	Behavior Modeling	Presence, realism
VR Classrooms	Adaptive Content	Engaged learning

IV. Findings and Challenges

A number of important conclusions are drawn from the current developments and uses of AI in AR and VR, along with important issues that need to be resolved for broad adoption and responsible growth. As Table II shows, AI models employed on commercial platforms share common themes but differ in responsiveness, size, and purpose.

Key findings:

- **Enhanced Perception:** Real-time spatial understanding and user interaction in immersive environments are greatly improved by AI models, such as sophisticated architectures like *Vision Transformers (ViTs)* and effective detectors like *YOLOv5*. [7], [9].
- **Deep Personalization:** Through sophisticated behavior modeling and affective feedback mechanisms, AI enables

AR/VR systems to offer unprecedented levels of personalization [16], [18].
- **Democratized Content Creation:** *Generative AI (GenAI)* models, including *Generative Adversarial Networks (GANs)* and *Denoising Diffusion Models*, are democratizing content creation by enabling rapid synthesis of complex virtual assets and environments [6], [20], [22].

Challenges: Despite these promising findings, several critical challenges persist:

- **Latency:** Heavy AI models often struggle to keep up on mobile or edge devices, causing delays that wreck immersion and might even induce motion sickness. This bottleneck demands smarter algorithms and faster networks like 5G or edge computing to keep the magic alive.
- **Privacy:** Significant privacy concerns are raised by the continuous and widespread collection of data in AR / VR environments, including biometric information, behavioral patterns, and spatial mapping. [25], [27].
- **Bias:** AI models have the potential to reinforce and even magnify societal biases if they are trained on biased or unrepresentative datasets. [29].
- **Authenticity:** Concerns regarding false information and the legitimacy of virtual identities are raised by the growing complexity of generative AI outputs, which makes it more difficult to verify digital content. [30].

V. Ethical and Reproducible Systems

Ethics and reproducibility are the twin pillars that hold up trust in AI-powered AR/VR worlds. Transparency is not just academic nitpicking; it is the lifeline that ensures these immersive systems remain fair, accountable, and worthy of our belief.

A. Reproducibility and Transparency

For academic rigor and public trust, the *reproducibility* (obtaining the same results using the same methodology and data) and *replicability* (obtaining the same finding with new samples) of AI models in AR/VR are crucial [33]. The CCSE 2025 conference explicitly outlines the reproducibility requirements, demanding a complete description of algorithms or resources or methods and actively encouraging public datasets, code sharing and baseline comparisons [1]. This transparency facilitates independent auditing, bias detection, and verification of claims.

B. Bias and Fairness

Bias in AI models can lead to discriminatory outcomes and diminish the inclusivity of AR/VR experiences [29]. For instance, gesture recognition systems might perform poorly for diverse user groups if trained on biased datasets [36]. To mitigate this, XR systems must be rigorously benchmarked across diverse user populations. Mitigation strategies include "fairness-aware machine learning techniques, data pre-processing strategies, model auditing frameworks, and post-deployment monitoring tools" [29]. *Explainable AI (XAI)* tools are also essential for enhancing transparency and understanding how AI models make decisions, helping to identify and correct biased outcomes [29].

C. Deepfakes and Identity Theft

The advanced generative capabilities of AI in AR/VR raise significant concerns regarding *deepfakes* (hyper-realistic synthetic media created using AI) and identity theft [30]. AI-generated avatars and synthetic media can be highly realistic, making it challenging to distinguish between authentic and fabricated digital identities [31]. To counter this, state-of-the-art detection methodologies are being rapidly developed, often leveraging AI itself to identify subtle artifacts in generated content [32].

D. Legal and Regulatory Compliance

Compliance with current data protection laws, such as GDPR, is made more difficult by the ubiquitous data collection that AR/VR systems entail. By passively gathering biometric and environmental data, AR/VR systems have the potential to unintentionally gather a significant amount of private data. [25]. To safeguard user privacy, technical solutions include *Differential Privacy*, *Homomorphism Encryption*, and *Federated Learning* [27], [28]. By enabling decentralized model training or allowing computations on encrypted data without sharing raw data, these techniques guarantee user control and compliance.

VI. Conclusion

AI is the beating heart powering the next wave of immersive experiences, transforming how we see, feel, and create within AR and VR. From lightning fast perception to deeply personal experiences and democratized creativity, AI unlocks worlds once confined to imagination. But as we move boldly forward, we must confront privacy, bias, and authenticity challenges head on: championing openness, ethics, and collaboration to build XR experiences that are not just dazzling, but deeply human.

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